



## AQ2.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

**Level 1:**

1 point

Consider a square matrix  $A = \begin{bmatrix} -12 & 0 & 0 \\ 2 & 0 & -1 \\ -10 & 1 & 0 \end{bmatrix}$ . If

- $P = -1M_{11} - 2M_{12} + 0M_{13}$   $P = -1M_{11} - 2M_{12} + 0M_{13}$
- $Q = -2M_{21} + 0M_{22} + 1M_{23}$   $Q = -2M_{21} + 0M_{22} + 1M_{23}$
- $R = -1M_{31} - 0M_{32} + 1M_{33}$   $R = -1M_{31} - 0M_{32} + 1M_{33}$

where  $M_{ij}$  is the minor with respect to the  $(i, j)$ -th entry, then choose the set of correct options.

☐

$$\det(A) = -2\det(A) = -2.$$

☐

$$P \neq QP = Q.$$

☐

$$P \neq Q \neq RP = Q = R.$$

☐

$$P = Q = RP = Q = R.$$

☐

$$Q \neq RQ = R$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\det(A) = -2\det(A) = -2.$$

$$P = Q = RP = Q = R.$$

1 point

Suppose for a real  $3 \times 3 \times 3$  matrix  $AA$ , there exists a real  $3 \times 3 \times 3$  matrix  $PP$  such that  $D = PAP^{-1}$   $D = PAP^{-1}$  is a real  $3 \times 3 \times 3$  diagonal matrix. Choose the correct set of options.

☐

$$\det(A)\det(A) \text{ must be equal to } \det(P)\det(P).$$

☐

$$\det(A)\det(A) \text{ must be equal to } \det(D)\det(D).$$

☐

$$\text{The matrix } AA \text{ must be equal to } DD.$$

☐

If  $DD$  is the identity matrix of order 33, then  $AA$  must be the identity matrix of order 3

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\det(A)\det(A) \text{ must be equal to } \det(D)\det(D).$$

If  $DD$  is the identity matrix of order 33, then  $AA$  must be the identity matrix of order 3

1 point

Choose the set of correct options for square matrices of order 33.



☐

If  $AB = 0AB = 0$ , then one of the matrices between  $AA$  and  $BB$  must have determinant 00.

☐

If  $AB = 0AB = 0$ , then both the matrices  $AA$  and  $BB$  must have determinant 00.

☐

If  $AB = 3IAB = 3I$ , where  $II$  denotes the identity matrix of order 33, then  $\det A = \frac{3}{\det(B)}$ .  
 $\det A = \det(B)3$ .

☐

If  $AB = 3IAB = 3I$ , where  $II$  denotes the identity matrix of order 33, then  $\det A = \frac{9}{\det(B)}$ .  
 $\det A = \det(B)9$ .

☐

If  $AB = 3IAB = 3I$ , where  $II$  denotes the identity matrix of order 33, then  $\det A = \frac{27}{\det(B)}$ .  
 $\det A = \det(B)27$ .

☐

If  $AB = 3IAB = 3I$ , where  $II$  denotes the identity matrix of order 33, then both  $AA$  and  $BB$  must have non-zero determinant.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If  $AB = 0AB = 0$ , then one of the matrices between  $AA$  and  $BB$  must have determinant 00.

If  $AB = 3IAB = 3I$ , where  $II$  denotes the identity matrix of order 33, then  $\det A = \frac{27}{\det(B)}$ .  
 $\det A = \det(B)27$ .

If  $AB = 3IAB = 3I$ , where  $II$  denotes the identity matrix of order 33, then both  $AA$  and  $BB$  must have non-zero determinant.

1 point

Consider a matrix  $A = \begin{bmatrix} abc \\ bca \\ cab \end{bmatrix}$ . If  $a + b + c$  is divisible by 66 then choose the

set of correct options.

☐

$\det(A)\det(A)$  is divisible by 55.

☐

$\det(A)\det(A)$  is divisible by 66.

☐

$\det(A)\det(A)$  is divisible by 22.

☐

$\det(A)\det(A)$  is divisible by 33.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\det(A)\det(A)$  is divisible by 66.

$\det(A)\det(A)$  is divisible by 22.

$\det(A)\det(A)$  is divisible by 33.

Let  $A = [a_{ij}]$  be a square matrix of order 3, where  $a_{ij} = iajj = i$ . Find  $\det(A)\det(A)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

**Level 2:**

Let  $A = [\delta_{ij}]$  be a square matrix of order 33, where  $\delta_{ij} = i \times j$ . Find  $\det(A)$ .

**No, the answer is incorrect.****Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

1 point

Consider the following square matrices:

$$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$$

$$A = \begin{bmatrix} 2013 & 2016 & 2019 \\ 2014 & 2017 & 2020 \\ 2015 & 2022 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2013 & 2019 \\ 2017 & 2014 & 2020 \\ 2022 & 2015 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 2013 & 2019 \\ 4034 & 2014 & 2020 \\ 4044 & 2015 & 2021 \end{bmatrix}$$

Choose the set of correct options.

☐

$$\det(B) = \det(A) \det(B) = \det(A).$$

☐

$$\det(A) = -\det(B) \det(A) = -\det(B).$$

☐

$$\det(C) \neq -2\det(B) \det(C) = -2\det(B).$$

☐

$$\det(C) = -2\det(A) \det(C) = -2\det(A).$$

**No, the answer is incorrect.****Score: 0****Accepted Answers:**

$$\det(A) = -\det(B) \det(A) = -\det(B).$$

$$\det(C) \neq -2\det(B) \det(C) = -2\det(B).$$

$$\det(C) = -2\det(A) \det(C) = -2\det(A).$$

**(Based on the below information answer questions 8 and 9).**Let  $A$  be a real  $3 \times 3$  matrix as given below:

$$A = [C_1 \ C_2 \ C_3]$$

where  $C_i$  represents the  $i$ -th column of the matrix  $A$ . Now let us consider the following set of matrices.

- $A_1 = [C_1 \ C_2 + 5C_3 \ C_3]$   $A_1 = [C_1 C_2 + 5C_3 C_3]$
- $A_2 = [C_1 + C_2 + C_3 \ C_2 \ C_3]$   $A_2 = [C_1 + C_2 + C_3 C_2 C_3]$
- $A_3 = [C_1 \ C_2 + 5C_3 \ 0]$   $A_3 = [C_1 C_2 + 5C_3 0]$
- $A_4 = [C_1 + C_2 \ C_2 + C_3 \ C_3 + C_1]$   $A_4 = [C_1 + C_2 C_2 + C_3 C_3 + C_1]$

1 point

Which of the matrices have the same determinant as that of  $A$ ?

☐

$A_1$

☐

$A_2$

☐

$A_3$

☐

$A_4$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$A_1$

$A_2$

1 point

Choose the set of correct options.

☐

$\det(A_3)$  cannot be determined from the given information.

☐

$\det(A_3) = 0$

☐

$\det(A_4) = 2 \det(A)$

☐

$\det(A_2) = 3 \det(A)$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\det(A_3) = 0$

$\det(A_4) = 2 \det(A)$

If the diagonal entries of a  $3 \times 3$  lower triangular matrix  $A$  are 1, 2, and 3, then find the sum of roots of the equation  $\det(A - xI) = 0$ , where  $I$  is the identity matrix of order 3.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 6

1 point

[Check Answers](#)

Your score is: 0/10